

SLA-backed network slicing for mission-critical radio and cellular services in airport operations, ensuring reliable communications for ground crew and safety-critical functions.

Transportation and Logistics × Tactical and mission-critical communications × Network slicing · OS143 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
67 is the world moving	39 can we play, can we win	MEDIUM 0 named in the evidence	€2.4bn partial evidence · per year	NOW budgeted procurement within 1...	L1 15 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

We propose SLA-backed network slicing to guarantee reliable, mission-critical radio and cellular communications for airport ground crews and safety functions. This opportunity is immediate because airports like Heathrow are actively seeking specialist support for their critical communications infrastructure, and network slicing technology has matured to meet stringent QoS demands.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€3.8bn €831m – €22.2bn	€2.4bn €518m – €13.8bn	€9.5m €2.1m – €55.2m	partial
Observed: tendered contract value, annualised	€115m	€115m	€459k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Network slicing	32.5 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.4 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Network slicing: No slicing series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (GB); the estimate covers the European footprint only.

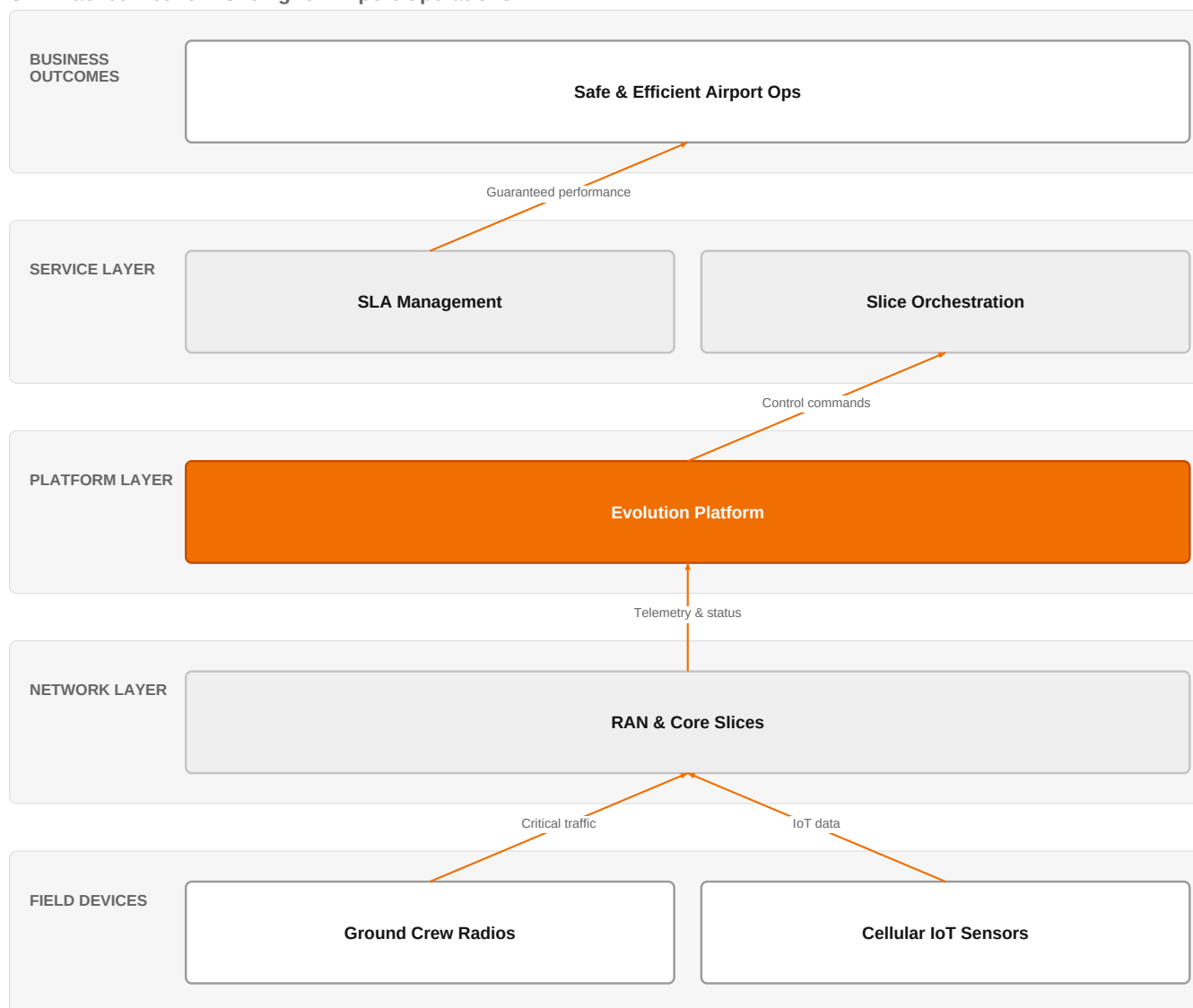
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	0.4 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (GB); the estimate covers the European footprint only.

The solution, in one picture

SLA-Backed Network Slicing for Airport Operations



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's Evolution Platform orchestrates network slices to guarantee SLA-backed communications for airport ground crews and safety functions.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Airports are facing increasing operational demands for reliable, low-latency communications, as evidenced by Heathrow's need for senior specialists in mission-critical radio and cellular systems. Network slicing has evolved to support ultra-reliable low-latency communications (URLLC) and enhanced mobile broadband (eMBB) in 5G and beyond, with research highlighting its role in meeting stringent QoS for mission-critical services. SLAs are recognized as essential for defining and managing QoS expectations, providing a contractual basis for guaranteeing performance.

Sources: SIG-01D38D7333B8 [contractsfinder.servic](#) 2026-08-12; SIG-C0BAE4D32A34 [openalex.org](#) 2025-08-19; SIG-AEA38615C065 [openalex.org](#) 2026-03-26

Who buys, and why now

The head of airport operations or the chief technology officer signs, but the pain is felt by the communications and safety teams who manage ground crew coordination and safety-critical functions. They are triggered by incidents of

communication failure during peak operations, or by audits revealing that current systems cannot guarantee the required reliability. The operational pain is dropped calls, delayed responses, and inability to prioritize safety-critical traffic over non-critical data.

Why it is hot now — every claim bound to a dated source

- SITA is hiring for mission-critical radio and cellular systems support at Heathrow Airport, indicating operational needs for critical communications. [SIG-01D38D7333B8]
- Mission-critical services in 4G/5G require stringent QoS, which network slicing can provide. [SIG-C0BAE4D32A34]
- SLAs are crucial for defining QoS expectations in service delivery, supporting the need for SLA-backed slicing. [SIG-AEA38615C065]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its Evolution Platform as the orchestration layer for network slicing, supported by its IoT, Digital, and Defense and security experts to design and deploy the service. The engagement would start with a discovery phase to map the airport's critical communication flows, then design a slice architecture with SLA guarantees, and finally implement and operate the solution with continuous optimization. The Evolution Platform would manage the lifecycle of slices, ensuring performance against SLAs.

Why Orange can win

Orange can win because it brings a combination of a carrier-grade platform (Evolution Platform) and deep expertise in IoT, digital, and defense/security, which is rare among competitors. The reference to Point One Navigation demonstrates Orange's capability in precise, reliable positioning, which is relevant to airport operations. However, the assets are thin—there is no specific airport or mission-critical communications reference, so Orange must prove its credibility through the platform's robustness and the team's expertise.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Evolution Platform	offer	L1	offer provides the technology in a matching domain	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.6; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include telecom operators like Colt, Deutsche Telekom/T-Systems, and Vodafone Business, who all sell network slicing and have strong connectivity offerings. Network equipment vendors such as Ericsson, Huawei, Nokia, and Boldyn Networks also compete, with deep expertise in RAN and slicing technology. Dedrone by Axon is a security specialist active in transportation, focusing on drone detection, which could be a complementary rather than direct competitor. The customer will compare Orange's ability to deliver a fully managed, SLA-backed service against these players' existing relationships and technology portfolios.

Sources: SIG-01D38D7333B8 contractsfinder.servic 2026-08-12

Competitor	Category	Position here	Basis	Evidence
Colt Technology Services	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions	structural	—

Deutsche Telekom / T-Systems	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions	structural	—
Vodafone Business	Telecom operator peer	sells Network slicing; active in Transportation and Logistics; competes in Connectivity Solutions	structural	—
Boldyn Networks	Network equipment vendor	sells Network slicing; active in Transportation and Logistics; competes in Connectivity Solutions	structural	—
Ericsson	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions	structural	—
Huawei	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Network slicing; active in Transportation and Logistics; competes in Connectivity Solutions	structural	—
Dedrone by Axon	Security specialist	active in Transportation and Logistics	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently prioritize traffic for ground crew communications over passenger Wi-Fi on your cellular network?
- 2 What is your current SLA for mission-critical radio and cellular services, and how often do you miss it?
- 3 Have you experienced any incidents in the last year where communication delays or failures impacted safety or operations?
- 4 Are you planning to upgrade your airport's communications infrastructure to 5G in the next 12 months?
- 5 Who is your current managed service provider for critical communications, and what are the pain points with them?

Objections you will meet

They will say	You can answer
We already have a reliable network from our existing provider; why do we need network slicing?	That's fair—your current network may be reliable for today's needs, but network slicing allows you to guarantee performance for critical services even during peak loads or emergencies, which a best-effort network cannot. It also gives you the flexibility to add new services without impacting existing ones.
Network slicing sounds complex and risky for mission-critical operations.	We understand the concern. That's why we propose a phased approach, starting with a pilot on non-critical services to prove the technology, and then gradually extending to mission-critical functions. Our Evolution Platform provides centralized management and monitoring, reducing complexity and risk.
We have budget constraints; this seems like a significant investment.	We hear you. However, the cost of downtime or communication failure can be much higher. By guaranteeing SLAs, we help you avoid operational disruptions and potential safety incidents. We can also structure the engagement to align with your budget cycles.

Risks and unknowns

The main risk is that airports may already have long-term contracts with existing telecom providers, making it hard to displace them. There is also uncertainty about the airport's willingness to adopt a new slicing solution given the critical nature of their communications. Additionally, the technical integration with legacy radio systems is not fully known, and the regulatory environment for spectrum and slicing in aviation is still evolving.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European airport operator's mission-critical communications tender, differentiating the Evolution Platform (L1) by highlighting its ability to deliver SLA-backed network slicing, and include the Point One Navigation reference as proof of precision technology integration.
Sales	In a conversation with a European airport operator's head of ground operations, say: 'We have a platform that can help you guarantee the reliability of your ground crew communications with network slicing, backed by SLAs—can we explore how that fits your operations?'

Strategist / Innovator	Commission a deep-dive brief on how the Evolution Platform (L1) can be positioned for SLA-backed network slicing in airport mission-critical communications, leveraging Orange Group researchers and Defense and security experts to validate the technical and regulatory angle.
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Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-01D38D7333B8	2026-08-12	contractsfinder.servi ce.g...	1	Resource Requirement - Senior Specialist/Engineer Service Operations for Mission Critical Radio...
SIG-A077923A1C80	2026-07-10	Springer Science and Busi...	1	ADONIS: Adaptive Digital Twin-Driven Orchestration for Network-Intelligent Slicing in 6G Edge E...
SIG-D8CAA774FDC4	2026-07-01	Next Research	1	Securing 5G network slicing for the smart grid network
SIG-AEA38615C065	2026-03-26	openalex.org	1	Service Level Agreement (SLA) and Security SLA (SecSLA): A Comprehensive Survey
SIG-40662980C98A	2026-03-20	openalex.org	1	NASP: Network slice as a service platform for 5G Networks
SIG-BE25FAACD322	2026-03-02	openalex.org	1	Bi-Objective Optimization for Scalable Resource Scheduling in Dense IoT Deployments via 5G Netw...
SIG-88237402830B	2025-11-28	SN Computer Science	1	Optimizing 5G Network Slicing for Heterogeneous Services
SIG-A70DE70DC023	2025-11-17	openalex.org	1	Deep Reinforcement Learning for Dynamic Network Slicing and Resource Orchestration in Software-...
SIG-263E4357EB27	2025-11-08	openalex.org	1	Investigating 5G Network Slicing Security Vulnerabilities Using Artificial Intelligence–Driven ...
SIG-2AE1E14EE912	2025-10-27	Institute of Electrical a...	1	ARCANE: Adversarial Resilience and Adaptive Network Slicing for UAV-based MEC
SIG-C0BAE4D32A34	2025-08-19	openalex.org	1	Mission-Critical Services in 4G/5G and Beyond: Standardization, Key Challenges, and Future Pers...
SIG-DD814ADD17F5	2025-01-01	cordis.europa.eu	1	Enabling Autonomous Aerial Systems through Operational Methods and Technological Enhancements
SIG-9FCAF3E022B5	2026-06-30	ASIS International	2	Drone Reports Near U.S. Airports Reflect Rising Issue for Critical Infrastructure - ASIS Intern...
SIG-5B36C4133AD6	2026-08-03	arxiv.org	3	Cross-Layer Optimization and System-Level Design of Next-Generation Wireless Networks via Intel...
SIG-7D18D409E4F0	2026-05-28	arxiv.org	3	Jamming-Resilient PRB Reservation for Latency-Critical O-RAN Network Slicing

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:25+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.